Design And Analysis of Algorithms
Lecture 16: Mirror Descent

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1 Mirror Descent: designed for ℓ_1 rather than ℓ_2

In this lecture, we're going to design an algorithm that in many ways is a lot like gradient descent, but will resolve the issue that we saw last time with gradient descent: namely, that its rate of convergence for the the n expert problem is $\Omega(n/\epsilon^2)$ rather than Hedge's $O(\ln n/\epsilon^2)$. The reason for this was that gradient descent wasn't designed with respect to the correct norm.

2 Background: Norms and Dual Normas and Spaces

We begin by defining a number of things which will help us define this improved algorithm. We begin with the notion of a norm, or measurement of vectors, in $\mathbb{R}^n$.

Definition 1 (Norm)**.** *A function* $||\cdot|| : \mathbb{R}^n \to \mathbb{R}$ *is a* norm *over* $\mathbb{R}^n$ *if*

- $||\vec{0}|| = 0$
- $||\alpha x|| = |\alpha|||x||$ *for all* $\alpha \in \mathbb{R}, x \in \mathbb{R}^n$.
- $||x + y|| \leq ||x|| + ||y||$ *for all* $x, y \in \mathbb{R}^n$ *(generalized Cauchy-Schwartz).*

You are likely familiar with a bunch of norms, but here are a few examples:

- $\ell_1(x) = \sum_{i \leq n} |x_i|$
- $\ell_p(x) = \left(\sum_{i \leq n} |x_i|^p\right)^{1/p}$
- $\ell_\infty(x) = \max_{i \in [n]} |x_i|$

We will also use the *dual* of a norm, or the dual norm. Informally, the dual norm of x measures the largest magnitude dot product between x and some y with norm at most 1:

Definition 2. *Given a norm* $||\cdot||$*, its dual norm is denoted* $||\cdot||_* : \mathbb{R}^n \to \mathbb{R}$*, and is defined as*

$$||x||_* = \sup_{y:||y|| \leq 1} |y^T x|.$$

Remark 1. *For finite dimensional spaces,* $(||\cdot||_*)_* = ||\cdot||_*$.

Some example norms and their duals include: $(\ell_1)_* = \ell_\infty$, $(\ell_2)_* = \ell_2$, and $(\ell_p)_* = \ell_{\frac{p}{p-1}}$. A useful fact about the dual of a norm is below, which allows us to move between considering inner products and their norms and dual norms.

Lemma 1. *For all x, y, we have that $\langle x, y\rangle \leq ||x||_*||y||$ via Cauchy-Schwartz.*

We will also use the concept of the dual of a convex set.

Definition 3. *Given a convex set $K \subseteq \mathbb{R}^n$, define the dual of K to be*

$$K* = \{y : x^T y \leq 1 \forall x \in K\}.$$

We finally note that it is natural to define Lipschitzness with respect to an arbitrary norm in terms of a function's gradient in the dual norm.

Definition 4. *We say that f is G-Lipschitz with respect to $||\cdot||$ if, for all $x \in \mathbb{R}^n$,*

$$||\nabla f(x)||_* \leq G.$$

3 "Fixing" gradient descent

We now have the basic tools we need to describe how to improve upon gradient descent. Recall that we were always considering loss functions f^t which were G-Lipschitz, continuous, differentiable, and convex. Also recall that gradient descent selects points $x^t \in K$, then defines $\tilde{x}^{t+1} = x^t - \eta^t \nabla f^t(x^t)$, then projecting $\tilde{x}^{t+1}$ back into K: $x^{t+1} = \arg\min_{y \in K} ||\tilde{x}^{t+1} - y||_2$.

Note that the gradient function at any point is some linear mapping describing the directional rate of change of f; it is thus a linear mapping over $\mathbb{R}^n$ and therefore a dual vector. It is therefore more natural to take a step opposite the gradient in the *dual* space of K rather than in K itself. So, we start in the primal space with some point $x^t \in K$, map x^t to the dual space, take a gradient step in the dual space, and then map back to the primal space, then back into the feasible region K. We need to define each of these maps to analyze the performance of such an algorithm. What properties should these maps satisfy/

Consider some function $h : \mathbb{R}^n \to \mathbb{R}^n$. Let h be continuous, α-strongly, and differentiable. Then, $\nabla h(x)$ will be a mapping from the primal to dual space, and we can define

$$y^{t+1} = y^t - \eta^t \nabla f^t(x^t) = \nabla h(x^t) - \eta^t \nabla f^t(x^t).$$

Then, let $\tilde{x}^{t+1} = (\nabla h)^{-1}(y^{t+1})$. We then project $\tilde{x}^{t+1}$ back into K in the event it is not already in K, by setting $x^{t+1} = \arg\max_{x \in K} D_h(x||\tilde{x}^{t+1})$, where D_h is the Bregman-divergence:

Definition 5. *Given a function h, let $D_h(x||y) = h(x) - h(y) - \langle \nabla h, y - x\rangle$. This is called the Bregman divergence between x and y, and it quantifies the slack in the convexity constraint of h between x and y.*

We can now describe Mirror Descent in its entirety as a function of h:

Algorithm 1: Mirror Descent

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for t = 1 to T do
    Let y^t = ∇h(x^t);
    Let y^{t+1} = y^t − η^t ∇f^t(x^t)(x^t);
    Let x̃^{t+1} = (∇h)^{-1}(y^{t+1});
    Let x^{t+1} = arg min_{x∈K} D_h(x||x̃^{t+1});
end
```

The performance guarantee for Mirror Descent can be written as follows.

Theorem 1. *Let h be α-strongly convex with respect to $||\cdot||$. Then, after T steps of Mirror Descent, when f^t is G-Lipschitz and convex,*

$$\sum_t f^t(x^t) \leq \sum_t f^t(x^*) + \frac{D_h(x^*||x_0)}{\eta} + \frac{\eta}{2\alpha} \sum_t ||\nabla f^t(x^t)||_*^2.$$

Before we prove this theorem, we consider what this means for two natural choices of norms and h's.

Example 1. *Let $||\cdot|| = \ell_2$ and $h(x) = \frac{1}{2}||x||_2^2$. Then, $\nabla h(x) = x$ and $D_h(x||y) = \frac{1}{2}||x-y||_2^2$. So the regret guarantee for Mirror Descent in this case is*

$$\frac{D_h(x^*||x_0)}{2\eta} + \frac{\eta}{2}\sum_t G^2 = \frac{||x^* - x_0||_2^2}{2\eta} + \frac{\eta}{2}\sum_t G^2$$

as we proved last lecture.

Example 2. *Let $||\cdot|| = \ell_1$ and $h(x) = \sum_i x_i \ln x_i$, which is $\frac{1}{\ln 2}$-strongly convex, for which $\nabla h(x)_i = (1 + \ln x_i)$, and $(\nabla h)^{-1}(y)_i = e^{y_i - 1}$. This should look familiarly like Hedge.*

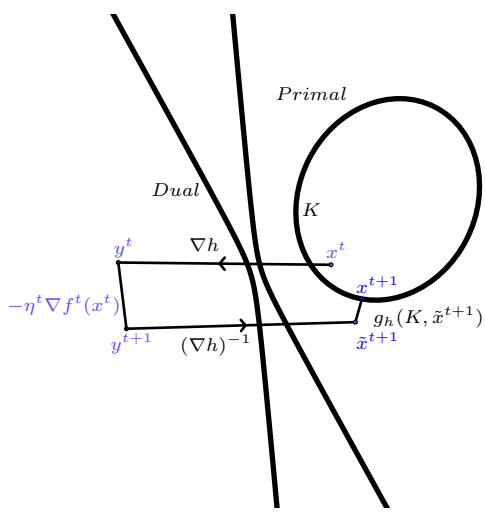
Then,

$$D_h(y||x) = -\sum_i y_i + \sum_i x_i + \sum_i x_i \ln \frac{y_i}{x_i} = -\sum_i x_i + \sum_i y_i + KL(y||x).$$

The regret guarantee is then

$$\frac{KL(x^*||x_0)}{\eta} + \frac{\eta}{2\alpha}\sum_t ||\nabla f^t(x^t)||_\infty^2 \leq \frac{\ln n}{\eta} + \frac{\eta}{2\alpha}\sum_t ||\nabla f^t(x^t)||_\infty^2$$

which in the experts setting, has $\nabla f^t(x^t) = \ell^t \in [-1,1]^N$, with $\ell_\infty(\ell^t) = 1$, so this becomes $\frac{\ln n}{\eta} + O(\eta T)$. This is quite similar to Hedge for this setting.



We now prove theorem 1.

Proof. As in the proofs for gradient descent, we define a potential function $\Phi^t = \frac{D_h(x^*||x^{t+1})}{\eta}$. The difference in potentials from time t to $t+1$ is thus

$$\begin{aligned}
\eta\left(\Phi^{t+1} - \Phi^t\right) &= D_h(x^*||x^{t+1}) - D_h(x^*||x^t) \\
&= h(x^t) - h(x^{t+1}) - \langle \nabla h(x^{t+1}), x^* - x^{t+1}\rangle + \langle \nabla h(x^t), x^* - x^t\rangle \\
&:= h(x^t) - h(x^{t+1}) - \langle y^{t+1}, x^* - x^{t+1}\rangle + \langle y^t, x^* - x^t\rangle \\
&= h(x^t) - h(x^{t+1}) - \langle y^t - \eta\nabla f^t(x^t), x^* - x^{t+1}\rangle + \langle y^t, x^* - x^t\rangle \\
&= h(x^t) - h(x^{t+1}) + \langle y^t, x^{t+1}\rangle + \langle \eta\nabla f^t(x^t), x^* - x^{t+1}\rangle - \langle y^t, x^t\rangle \\
&= h(x^t) - h(x^{t+1}) + \langle y^t, x^{t+1} - x^t\rangle + \langle \eta\nabla f^t(x^t), x^* - x^{t+1}\rangle \\
&\leq \frac{\alpha}{2}||x^{t+1} - x^t||^2 + \langle \eta\nabla f^t(x^t), x^* - x^{t+1}\rangle && \text{By } \alpha\text{-strong convexity of } h.
\end{aligned}$$

Then, we can do the calculation

$$
\begin{aligned}
\sum_t f^t(x^t) - \sum_t f^t(x^*) &= \sum_t \left(f^t(x^t) - f^t(x^*) + \Phi^{t+1} - \Phi^t \right) + \Phi^0 - \Phi^T \\
&\leq \sum_t \left(f^t(x^t) - f^t(x^*) + \frac{1}{\eta} \left(\frac{\alpha}{2} ||x^{t+1} - x^t||^2 + \langle \eta \nabla f^t(x^t), x^* - x^{t+1} \rangle \right) \right) + \Phi^0 - \Phi^T \\
&= \sum_t \left(f^t(x^t) - f^t(x^*) + \left(\frac{\alpha}{2\eta} ||x^{t+1} - x^t||^2 + \langle \nabla f^t(x^t), x^* - x^{t+1} \rangle \right) \right) + \Phi^0 - \Phi^T \\
&\leq \sum_t \left(f^t(x^t) - f^t(x^*) + \left(\frac{\alpha}{2\eta} ||x^{t+1} - x^t||^2 + \langle \nabla f^t(x^t), x^* - x^t \rangle + \langle \nabla f^t(x^t), x^t - x^{t+1} \rangle \right) \right) + \Phi^0 - \Phi^T \\
&\leq \sum_t \left(\frac{\alpha}{2\eta} ||x^{t+1} - x^t||^2 + \langle \nabla f^t(x^t), x^t - x^{t+1} \rangle \right) + \Phi^0 - \Phi^T \\
&\leq \sum_t \left(\frac{\alpha}{2\eta} ||x^{t+1} - x^t||^2 + ||\nabla f^t(x^t)||_* ||x^t - x^{t+1}|| \right) + \Phi^0 - \Phi^T \\
&\leq \sum_t \left(\frac{\alpha}{2\eta} ||x^{t+1} - x^t||^2 + \frac{1}{2} \left(\frac{\eta}{\alpha} ||\nabla f^t(x^t)||_*^2 + \frac{\alpha}{\eta} ||x^t - x^{t+1}||^2 \right) \right) + \Phi^0 - \Phi^T \\
&= \sum_t \frac{1}{2} \left(\frac{\eta}{\alpha} ||\nabla f^t(x^t)||_*^2 \right) + \Phi^0 - \Phi^T \\
&\leq \sum_t \frac{1}{2} \left(\frac{\eta}{\alpha} ||\nabla f^t(x^t)||_*^2 \right) + \Phi^0 \\
&= \frac{D_h(x^* || x^0)}{\eta} + \frac{\eta \sum_t ||\nabla f^t(x^t)||_*^2}{2\alpha}
\end{aligned}
$$

as desired. $\square$